

9. Memory Hierarchy & Caching Deep Dive

Computer Science Fundamentals | Module 3: Memory Subsystem | CS_Fundamental-00

Interview Memory Hook: "Jo memory jitni fast hogi, woh utni hi mehngi aur chhoti hogi. Jo memory sasti aur badi hogi, woh dheemi hogi." Is trade-off ko solve karne ke liye **Memory Hierarchy** aur **Multi-Level Cache** bani!

9.1 Memory Hierarchy Pyramid & Core Trade-offs

- **Level 1 (Top) - CPU Registers:** Fastest (0.25-0.5 ns), small (few KB), expensive.
- **Level 2 - Cache Memory (SRAM):** L1, L2, L3 caches (1-20 ns).
- **Level 3 - Main Memory (DRAM):** Primary RAM (50-100 ns, 8-64 GB), volatile.
- **Level 4 - Secondary Storage:** NVMe / SSD / HDD (Microseconds to milliseconds, Terabytes).
- **Level 5 (Bottom) - Magnetic Disks & Cold Storage:** Backup / Tapes.

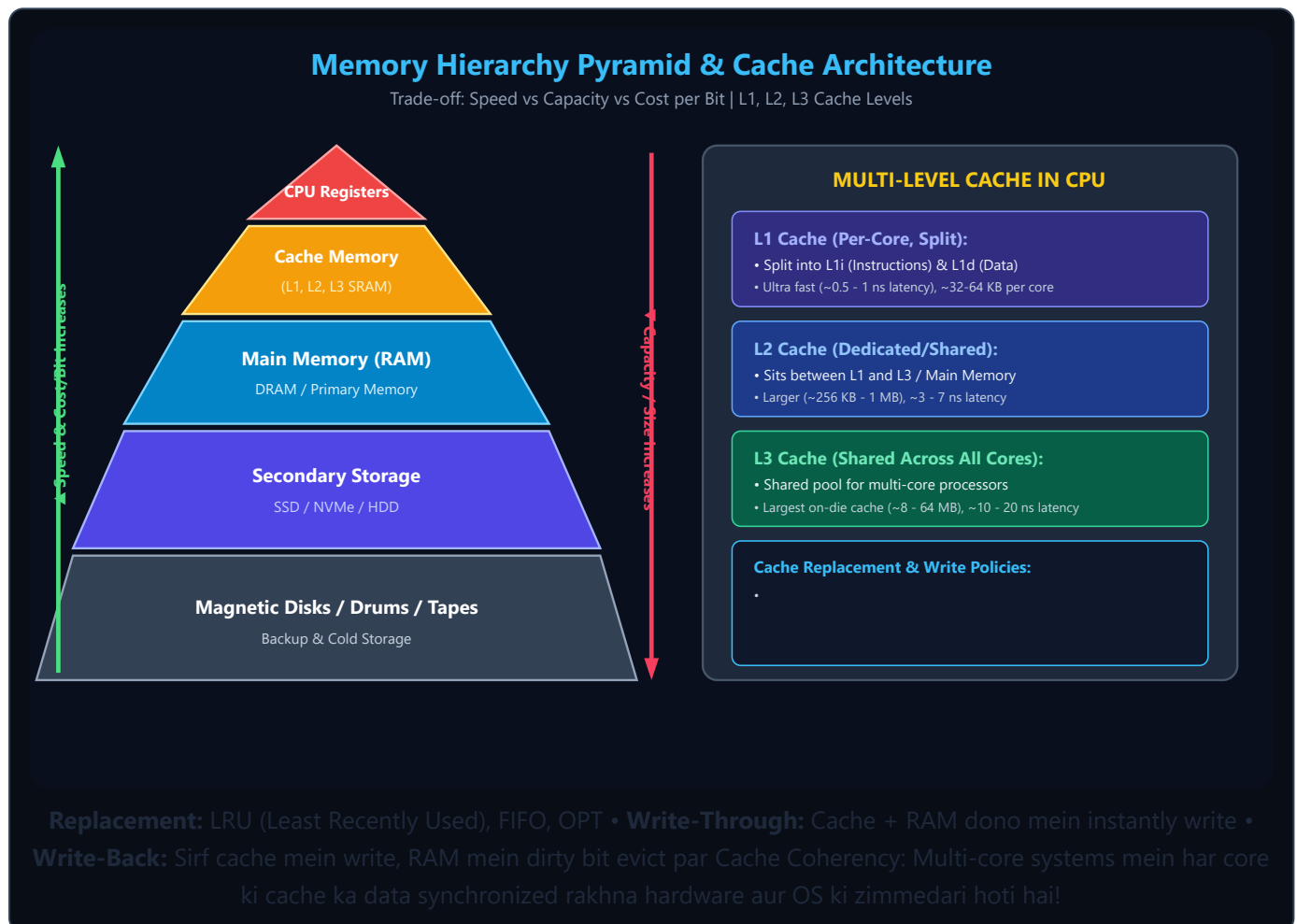


Fig 9.1: Memory Hierarchy Pyramid & Multi-Level Cache Architecture

9.2 Caching & Principle of Locality

- **Temporal Locality:** Jo data abhi access hua hai, woh jald hi dobara access hoga (Loops, counters).

- **Spatial Locality:** Jo memory address access hua hai, uske agal-bagal ke addresses bhi jald access honge (Arrays, sequential code).
- **Average Memory Access Time (AMAT):** $AMAT = T_{cache} + (1 - H) * T_{RAM}$

9.3 Multi-Level Caches (L1, L2, L3)

- **L1 Cache:** Per-core dedicated, split into L1i (Instructions) & L1d (Data). ~32-64 KB, ~1 ns.
- **L2 Cache:** Intermediate buffer between L1 and RAM. ~512 KB - 2 MB per core.
- **L3 Cache:** Shared across all CPU cores on die. ~8 - 64 MB, ~10-20 ns.

9.4 Cache Policies: Replacement & Writes

- **Replacement:** LRU (Least Recently Used), FIFO, Random, OPT.
- **Write-Through:** Cache aur RAM dono mein instantly write (consistent, slower).
- **Write-Back:** Sirf cache mein write, Dirty Bit set. RAM mein evict hone par likhta hai (fastest).
- **Cache Coherency:** Multi-core mein sabhi cores ke cache data ko synchronize rakhna (MESI protocol).